Antonino Pio Lupo

DATA SCIENTIST · SPORTS ANALYTICS

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PROFILE

Data Scientist working in football analytics, building end-to-end pipelines that turn public and proprietary match data into predictive models and decision-ready insight. Python, R and SQL across the full lifecycle: scraping and ETL into Azure, feature engineering, model calibration, and backtesting with Monte Carlo simulation. Currently an Analyst at Football Benchmark; MA in Data Science for Economics, University of Milan. Targeting Data Scientist and Data Engineer roles in the football industry.

EXPERIENCE

Analyst · Football Benchmark

Budapest · 02/2025 – Present

- Engineered end-to-end Python scraping pipelines to extract public football data and seamlessly integrate it into Azure databases for streamlined retrieval.
- Optimized and refactored machine learning workflows, enhancing the data lifecycle from initial preparation through to final predictions.
- Contributed to a Python pipeline hosted on Streamlit that automates presentation creation end-to-end, retrieving data from the Azure database, staging it in Excel, and auto-generating PowerPoint slides with populated charts.
- Collaborated in the refinement and testing of specific endpoints of the company's API.
- Supported the company's newsletter by analyzing socio-economic trends in football and designing insightful data visualizations using Python and Excel.

Data Analyst Intern · Aesys S.r.l.

Remote · 04/2025 – 06/2025

- Developed a sentiment analysis pipeline in Python to process 56,000+ tweets, utilizing NLP techniques to quantify and visualize communication tone shifts.
- Optimized Azure cloud ETL pipelines, restructuring data flows to improve processing scalability and reduce latency for large-scale datasets.
- Deployed interactive data visualization apps transforming raw unstructured data into strategic insights for executive decision-making.

Financial Analyst Intern · Banca Mediolanum S.p.A.

Bologna · 10/2022 – 12/2022

- Analyzed portfolio performance metrics to identify growth opportunities, directly supporting senior advisors in wealth management strategies.
- Conducted market research and financial reporting, providing data-driven recommendations for client asset allocation.

SELECTED PROJECTS

Master Thesis: Machine Learning Model for Shot Dominance

R · Python

- Developed a predictive model to forecast shot dominance in football matches, identifying key performance indicators.
- Engineered a robust data pipeline using R for web scraping and Python for cleaning, standardizing, and merging disparate data sources.
- Designed predictive features to enhance interpretability and calibrated multiple ML algorithms to optimize performance.

Betfair Value Betting Bot

Python · REST API

- Built an automated Python service that reads model probabilities from Excel, fetches live odds through the Betfair Exchange JSON-RPC API, and places back and hedging lay orders end-to-end.
- Implemented certificate-based non-interactive authentication, fuzzy team-name resolution with a persistent mapping layer, a dry-run execution mode, and automated CSV/Excel run reporting.

Reinforcement Learning Blackjack Agent

OpenAI Gym · Q-Learning

- Developed a custom OpenAI Gym environment to train a Q-Learning agent using Linear Function Approximation.
- Engineered state representations using binary encoding and polynomial features, achieving ~49% win rate over 1M episodes.

EDUCATION

MA in Data Science for Economics · University of Milan, Italy

09/2023 – 10/2025

Grade: 106/110. Focus: Machine Learning, NLP, Statistical Methods, Big Data Architecture and Security. Football Analytics lab with Soccerment: Random Forest regression model predicting match corner counts from event and tracking data.

BA in Business and Economics · University of Bologna, Italy

09/2020 – 09/2023

Grade: 110L/110 (Summa Cum Laude). Exchange program: Dickinson College, PA, USA (01/2023 – 05/2023).

TECHNICAL SKILLS & LANGUAGES

Core stack	Python, R, SQL, Git
Data eng & cloud	Azure Cloud, ETL processes, web scraping (Selenium, selenider, chromote), Streamlit, Power BI, Excel
Modeling	Machine learning (LightGBM, Random Forest), reinforcement learning, statistical analysis, NLP, Monte Carlo simulation, backtesting
Visualization	Matplotlib, Seaborn, Power BI
Languages	Italian (native), English (fluent — IELTS 7.5, C1), German (B1)